

Teamwork plus Technology Equals Reduced Emissions, Reduced Energy Usage, and Improved Productivity for an Oil Production Facility

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ABSTRACT

Suncor Energy Inc. developed a long term plan to expand production from its oil sands operation north of Fort McMurray, Alberta up to 500,000 to 550,000 barrels/day in 2010-2012, while reducing the per barrel energy usage, emissions, and long term production cost target to under C\$10 per barrel. Current production has been increased to 225,000 barrels/day through the application of advanced technologies and an ongoing series of expansion projects. This paper is limited to the integration of an advanced control system project into the existing Steepbank Mine and Extraction Process Plant. The Extraction Process separates the raw oil called bitumen from the mined ore and is the largest energy user on the site. A control system and energy audit was conducted in the fourth quarter of 2000 to form the basis for the ongoing energy reduction and production improvement Advanced Control Project. The project consists of initiatives ranging from simple operating procedures to integrating the new control system into major process modifications. When performance of the new Extraction control system controls was evaluated against the original 2000 baseline, the overall production gain was 9% to 12% valued at C\$44 million/year with an additional C\$8.7 million/year reduction in hot process water.. The Advanced Control Project is part of a greater Extraction production improvement and energy reduction program returning a total 2001 production increase of C\$104 million/year when evaluated against the 2000 baseline. At the end of 2002 the Extraction Unit team effort returned an additional production increase of C\$140 million/year while reducing hot process water energy usage an additional C\$2.5 million/year.

INTRODUCTION

Suncor Energy owns and operates the first commercially successful oil sands extraction facility in the world. Oil sands are unconsolidated oil sand deposits impregnated with petroleum, referred to as bitumen. Alberta has the world's largest deposit of oil sands estimated at 1.3 trillion barrels¹. Suncor has developed several processes to mine ore and produce high quality synthetic crude oil for refining at other facilities. The project focused on the new ore supply process train consisting of the Steepbank Mine, Plant 82 OPP and Extraction Unit, the largest user of thermal and electric energy at the facility. There was an urgent need to improve process train production to offset the closure of the existing mining lease 86/17. The project leveraged the efforts of internal teams, equipment vendors, and university research programs to assist ongoing capital projects in reaching the ROI targets.

PROCESS DESCRIPTION

Mining

The mining process has evolved from the original bucketwheel and conveyor mining to a more efficient shovel, truck, and hydrotransport method. This ultramodern mining complex uses several classes of shovels and a fleet of heavy hauler trucks to extract the ore from the mine



face and deliver it to the Ore Processing Plants

Figure

(OPP). Loaded trucks weigh over one million pounds and the size of equipment is shown in Figure 1. The shovels and trucks are equipped with computers, GPS, and telecommunication equipment to communicate with mine dispatch. The Mine Dispatcher's job resembles that of an air traffic controller as he orchestrates all the shovels and trucks to assure delivery of the proper tonnage and ore quality to the OPP's.

Ore Processing Plant (OPP)

A simplified process diagram is shown in Figure 2. The extraction process is a version of the Clark Hot Water Extraction (CHWE)² process modified to operate at lower temperatures and with a minimum of chemical additives. Ore is received and crushed by the sizer at the OPP and conveyed to the rotary breaker where it is mixed with hot water, collected in the breaker pump box, and pumped by four 2500HP pumps in series through the hydrotransport lines to the extraction plant.

Extraction

Extraction receives the ore from the hydrotransport and separates the bitumen from the tailings (sand and other residue). The bitumen then travels through the dearator,

interstage tank, froth treatment plant, and to the diluted bitumen storage tank. The tailings travel to the consolidated tailing system to facilitate water removal and is pumped to the tailings ponds to settle out the remaining solids. The pond water is then recycled to minimize overall plant water usage.

Upgrading

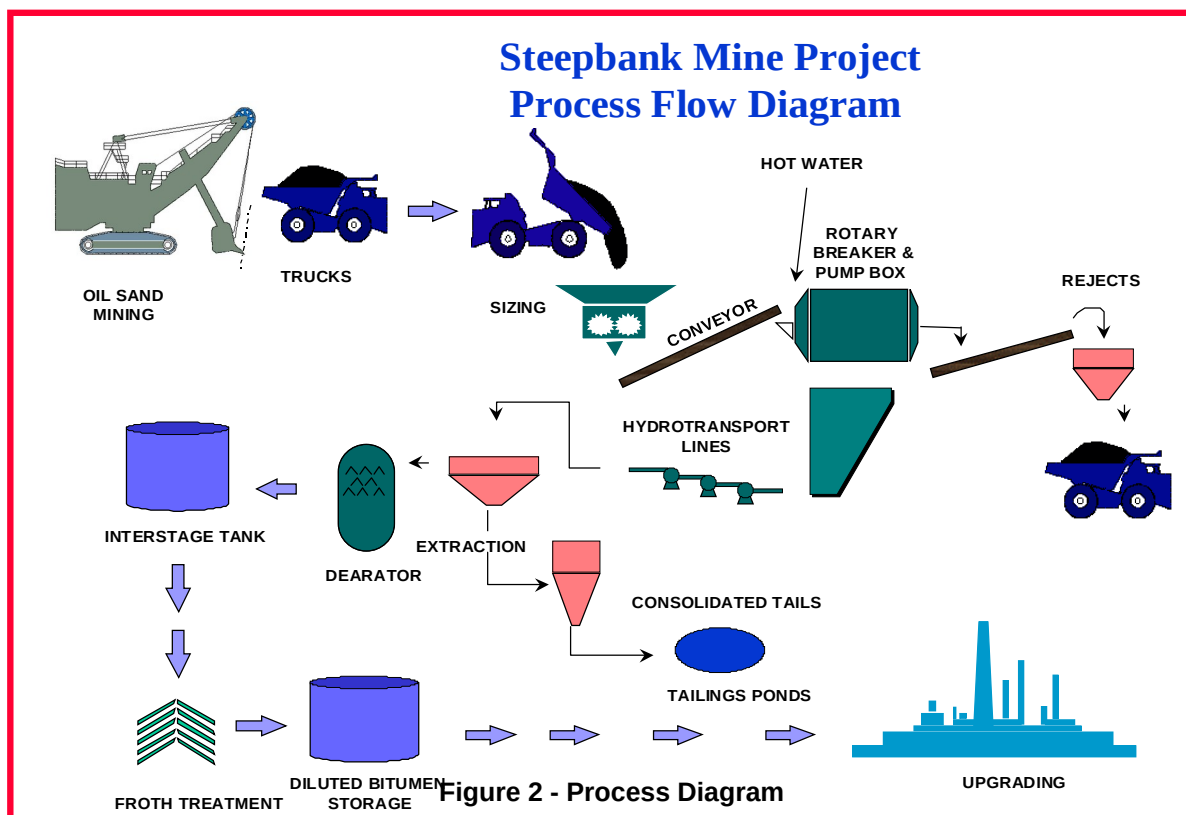
The final process unit is the Upgrader that converts bitumen to a high quality synthetic crude oil. The synthetic crude is transported to refineries for processing into petroleum products.

PROJECT METHODOLOGY

Suncor is undergoing an intensive capital expansion program that necessitated a review of the existing Extraction Process Unit. The initial requirements were an energy, process, and control system audit to identify opportunities for improvements. Upon completion of the audit, an Advanced Control System Project was initiated.

Project Audit

Process and control system audits take many different forms based on the client's



requirements. The methodology used in this audit was detailed in an earlier paper³ and targeted the needs defined by the client. The traditional approach for an effective audit contains the following steps:

- Project Definition: definition of the project objectives
- “As-Is” System Status: clearly define the system as it stands.
- Analysis: defined by the project objectives; develop Control Initiatives with savings
- Verify Findings: validate the findings and verify the results with the client.
- Issue a Report: finalize the system analysis based on project objectives and verification issues.

This audit was unique in two ways. First, Suncor was undergoing a fast track expansion and debottlenecking program to increase production capacity, reduce energy usage, reduce plant emissions, and improve product quality. Second, the employees at Suncor were empowered to make changes; their input was actively sought at all levels during the audit and execution process. The aggressive schedule required the use of informal high performance work groups to focus on areas of opportunities. The work groups consisted of representatives from operations, maintenance, project administration, process engineering, instrument and electrical technicians, control systems engineering, information technologies, and the training department. This free exchange of information allowed the team members to be more effective and focus on areas of new work while supporting the existing workload. The extraction control system developed the following:

- Control Initiatives were used to document the different projects that were discussed.
- Data Sources were expanded to include the Equipment Loss, Process Loss and Incident Reporting System to capture the history and cost of past incidents.
- Process data was obtained from the plant historian and control systems to provide an analysis of the plant control capabilities during steady state operation and process upsets.

Project Execution

A review of the audit team project initiatives indicated that improvements in production and density control through the hydrotransport system would result in returns that justified completing additional energy and cost avoidance initiatives. The teams working on the audit were moved to the project for the following reasons:

- Provide personnel familiar with the project to decrease project cycle time.
- Coordinate with ongoing multiple projects providing more “bang for the buck”.

Flexibility and resourcefulness is mandatory on a project with strict production schedules especially when it is necessary to modify existing systems. Due to the size and complexity of the original project, it is broken into several manageable phases shown in Figure 4. Phase 1 includes Process Train #2, which started at the mine and worked into the extraction separation facility. This allowed integrating the control modifications into a single line during process outages to permitted validation of the modifications before integration into Process Train #1. Phase 2 included integrating the new systems into Train #1 at the mine and continuing through the remaining portion of the extraction plant and into final tailings. This approach was modified because process changes and mandatory modifications rendered previous work unusable. An example was the loss of the critical apron feeder bin level measurement when a large grizzly was installed on top of the apron bin. The grizzly blocked the scanning laser causing a one-year schedule slippage while a platform and instrument system was evaluated and installed.

This schedule delay forced the development of the intermediate Phase 1 ½ for execution of the following systems:

- Operations requested the Train #2 control components be integrated into Train #1 for improved performance.
- An adaptive control for a new style of breaker pump box was developed.
- A new control strategy was developed for the removal of three hydrotransport tanks and closely coupling the breaker transfer and hydrotransport pumps.

- The mine interface was enhanced to include an ore delivery and ore quality display and reporting systems.
- A Video Event Recording Analyzer (VERA) was developed to monitor critical ore processing systems.

CONTROL STRUCTURE

The Suncor Energy Oil Sands plant integrates all aspects of the Mine and Extraction Plant Operation into a cohesive process management and reporting system. The control system architecture is shown in Figure 3 and provides a secure process control system architecture while presenting strategic operating information to the business management and reporting systems. The control system hierarchy levels are summarized below:

- The Process Connected equipment are devices such as transmitters, valves, variable speed drives, etc. that are used to measure and manipulate the process.
- The Distributed Control System (DCS) / Programmable Logic Control (PLC) provide the intelligence of the system to the Process Connected Equipment to read the process inputs and manipulate the process output devices.

- Several Control Rooms house the Man Machine Interface (MMI) equipment necessary to view and manipulate these systems. These control centers are located throughout the complex for the use of front line operation and supervisory personnel.

Recently, a series of Advanced Process Management Initiatives such as the Production Rate Control (PRC)[®], MineVu and constraint reporting systems were initiated to support the business objectives of the Oil Sands facility.

Plant wide scheduling and reporting systems are supported through the Information Technologies (IT) groups. These systems bridge the process control and information technology domains using firewalls and other security techniques to prevent unauthorized manipulation of the process variables.

Figure 3 – *Extraction Control System Architecture* identifies the subsystems completed to date with red check marks. As the production continues to improve, an ongoing effort

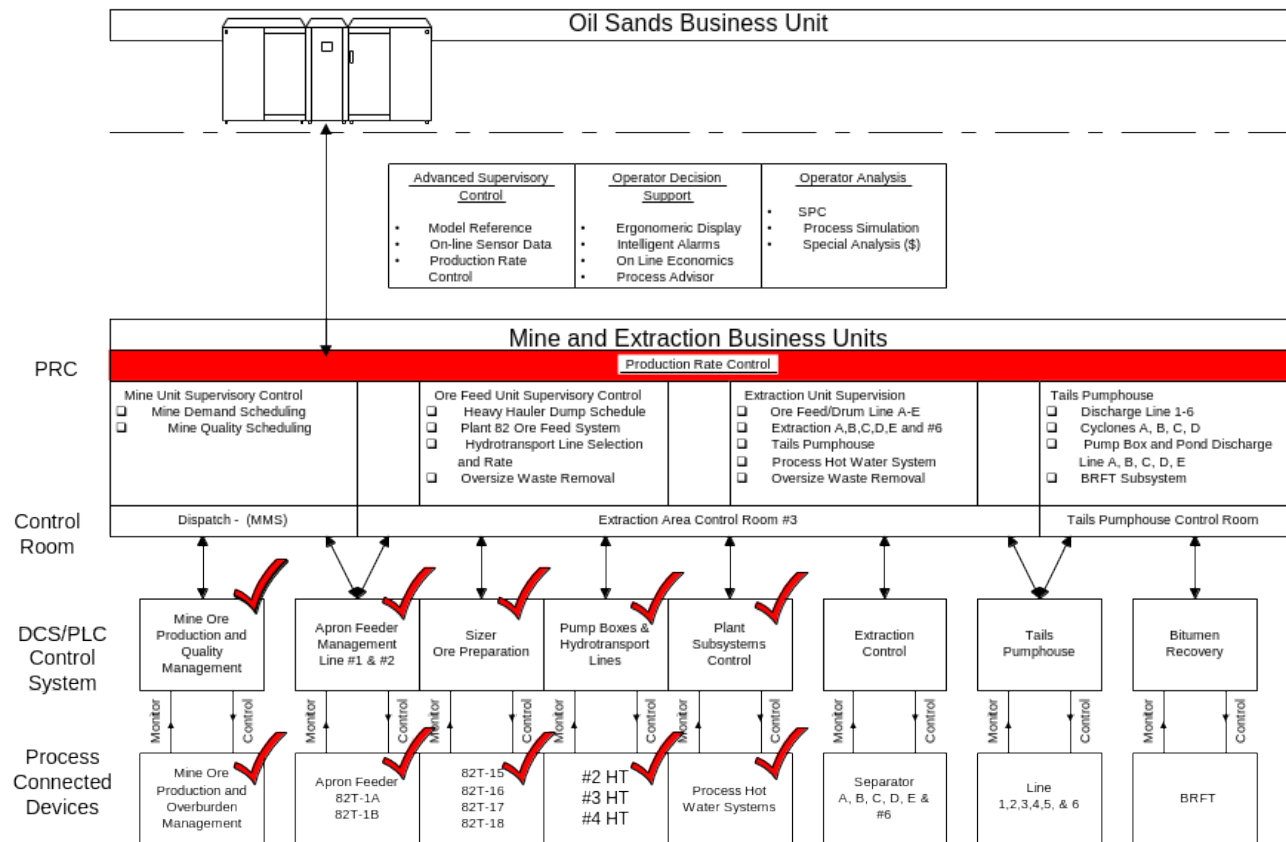


Figure 3 - Extraction Control System Architecture

continues to tweak the completed systems to improve performance.

Project Objectives

The audit indicated that substantial energy savings and production increases could be obtained through process modifications and control system enhancements. Due to the size and complexity of the Advanced Control System project, the effort was further broken into two broad categories, Foundation Work and Advanced Control System. The overall structure of the project is presented in Figure 3 and detailed below:

Foundation Work

The primary objective of the Foundation Work was to document the existing process and control systems and work with the various teams to correct control and process problems. This became a formidable task due to the lack of up-to-date information. The project teams performed the following tasks:

- Documentation validation was mandatory. The area teams walked down and documented the process and instrument diagrams and provided process control narratives of the programs and logics. This included updating process flow diagrams, flowcharting process control programs and the development of SAMA diagrams of the existing control strategies. The format was designed to be compatible with the efforts of a separate team that was developing training manuals.
- System Loading Study and Diagnostics was performed and a system loading study developed to assure overall system capacity and reliability.
- An overall system software cleanup was performed to remove inactive points and control programs.
- Inoperative control loops associated with the Advanced Control Platform were fixed. Once the overall system is operational, a systematic control loop evaluation and tuning will be accomplished.
- Several new technologies were integrated into the process to provide operations with the information they need to improve process controllability.

- Once the documentation was developed and organized, a mechanism will be developed to allow the DCS system to automatically update the system documentation. This assures the most current documentation to engineering, operations and the E&I groups.

Advanced Control System Work

As the foundation control system was documented and corrected, the implementation of advanced regulatory and supervisory control systems began. The specific objective was to find and apply the very best technology to the process and provide a seamless integration of the existing and new applications while minimizing operator training. The result was a series of new technologies listed below:

- MineVu a Mine Dispatch/Process Plant Distributed Control System interface to provide predictive tonnage available and anticipate ore quality to extraction. The objective of the system is to match the performance of both areas and always push to the available production capacity.
- Integration of the overall system into a Production Rate Controller (PRC[®]) to coordinate unit production rate. This allows extraction to move the process unit to match the available ore processing rate of the mine.
- Integration of new measurement and control technologies that provide more accurate and reliable control. This included parallel implementation of control initiatives by several teams.
- A constraint reporting system responding online and in real time to help operations make better decisions.
- Integration with the enterprise management and scheduling system to maximize equipment utilization.
- System diagnostics and reporting to allow proactive rather than reactive operations to abnormal operating conditions.

The approach was successful. However, coordinating a fixed schedule operating project with a series of research and development tasks increased project engineering in some areas by a factor of three. The benefit was that the resulting

systems provided operations, engineering, and maintenance exactly what was required.

CONTROL ALGORITHM

The overall Production Rate Control (PRC[®]) has been used in the pulp and paper industry since 1990 and is now being applied in the extraction process. The PRC[®] is detailed in Figure 4 below. The primary control strategy objective was to coordinate the operation of individual process units into one overall process area. As the project developed, regulatory control strategies were integrated into the blocks marked "Regulatory Control". CL Programs and Multivariable Constraint Controls were integrated into the block called "Process Unit Constraint Controller". The overall system control was coordinated through the block called "Production Rate Controller" supervisory system.

Another major objective of the control strategy was to drive the extraction production to the maximum production rates from the mine while minimizing the effects of ore starvation and ore grade problems. The MineVu system

interfaces with the existing mine management system to forecast the mine performance rates. The PRC system obtains data from MineVu system to match the extraction production rate with the mine ore delivery rate. This maximizes truck utilization and maximizes the extraction production rate while reducing the risk of ore starvation.

The new controls were implemented in the revised Phase 1 and 1 ½ portion of the project and are applied to Train #1 and #2. Completed systems are checked off.

As the project progresses into Phase 2 the balance of extraction process will be integrated into the PRC[®] to manipulate the feed rate to the maximum rate allowed by the entire process unit with FTPH restrictions feeding back to the OPP Feed Rate Control.

The plan was effective but there were several problems. Three tanks were demolished and one tank modified, which caused multiple changes to the control philosophies.

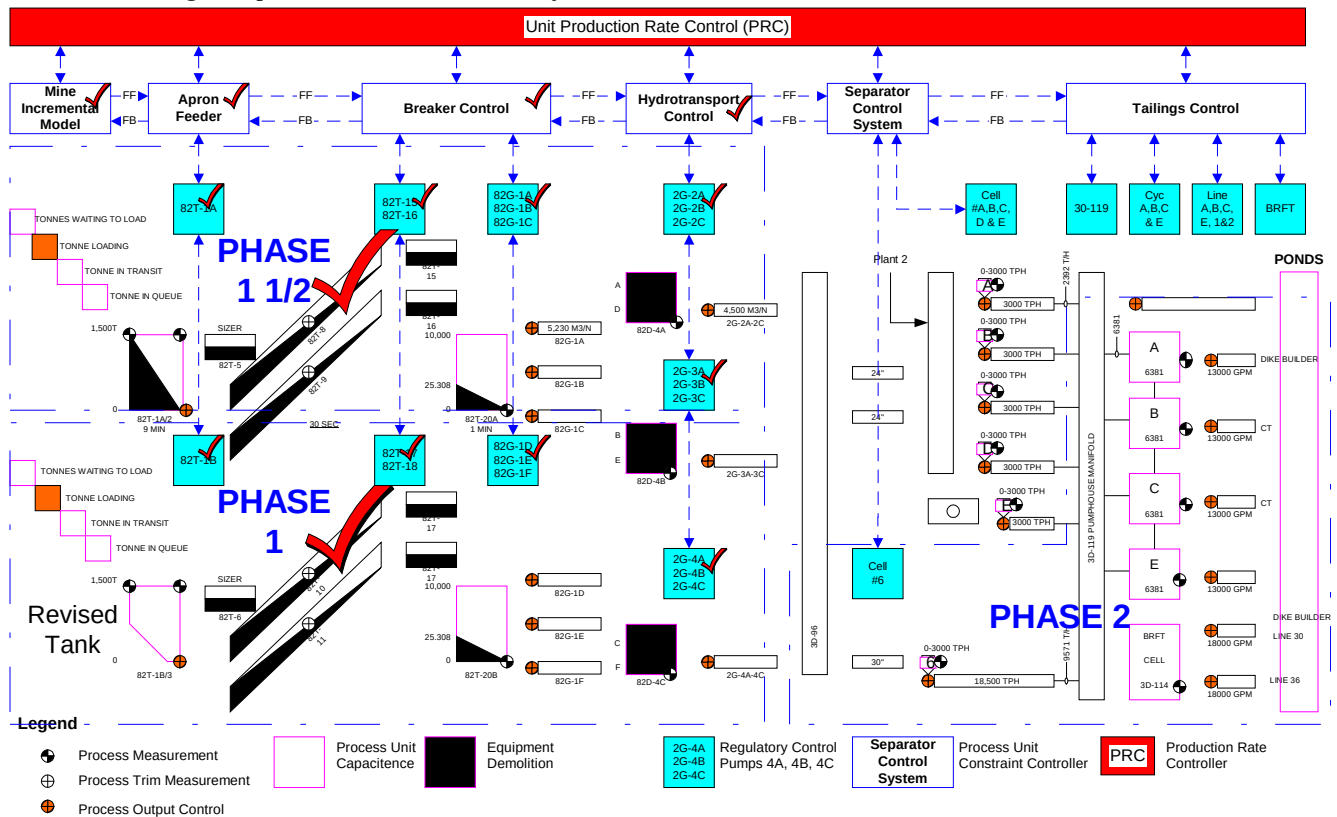


Figure 4 - Control System Algorithm and Scope of Work

ADVANCED PROCESS MANAGEMENT TECHNIQUE

During the course of the project, a tool was applied to validate the control system methods and techniques. A modeling technique of “Advanced Process Management” was implemented to capture the process and operational dynamics of the mine, extraction process, and control systems. This technique was discussed in detail in a previous paper⁴ and is used to reduce the cost of the control system development and reduce the risk of errors during the control system implementation. The technique uses a simplified process model that allows testing control system and operating strategies off line prior to implementation on the actual process.

A second mechanism was applied to facilitate data analysis and the prototyping of control strategies. An Advanced Control Platform (ACP) was implemented on the project. This platform provided a faster data historian (1 second) than the current Distributed Control System (DCS) and a tool set for faster development and testing of control strategies. This permitted interfacing to the existing control strategies in Direct Digital Control Mode for testing. Once these strategies were perfected, they would be translated into the DCS system. This opened capacity for a new system in the ACP and providing a consistent control strategy in the DCS.

RESULTS

A simple measurement target was developed in 2000 to focus the project team and provide a long-term performance index. The target was to

increase tonnage throughput of the hydrotransport system by one truck per hour [300 Tonnes] and improve density control to increase the average operating density of the slurry to 1.55 SG under normal operating conditions to reduce hot water usage. The results of the 2002 control system performance test is shown in Table 1 – *Early Project Performance Index* and credits the control system with an overall production gain of between 9% to 12% valued at C\$44 million/year while reducing hot process water energy usage by C\$8.7 million/year.

The three-year progress of the overall project is shown in the supporting histograms of production and density hourly average measurements in Figure 5a, 5b and 5c. When evaluated in 2001 against the 2000 baseline, the Extraction Unit team returned a production increase of C\$104 million/year. In 2002 the Extraction Unit team effort returned an additional production increase of C\$140 million/year while reducing hot process water energy usage an additional C\$2.5 million/year.

It is difficult to assign the value of project benefits when multiple capital and control projects occur simultaneously. The simplest method was to conduct a performance run with the controls turned off, versus the controls turned on. The winter period was selected to provide the most difficult operating period to apply two evaluation methods. The first test to determine long-term system availability and operator acceptance started in January and returned a 9% production improvement. The second test, displayed in Table 1, was a more controlled test over a one-month period and returned a 12% production improvements.

Description Advanced Control Evaluation 2002	Test Run 1/1 to 4/7 General Operation New Regulatory & Advanced		Operating Time	Production Contribution	Test Run 2/24 to 4/7 Pushed Advanced Control		Operating Time	Production Contribution
	Tonnes	Composite SG	Percentage	Percentage	Tonnes	Composite SG	Percentage	Percentage
Production Rates With New Regulatory Controls and without Water and PRC Programs								
Total Tonnes for Period	11,035,874		70%	68%	3,692,724		44%	41%
Average	6,804	1.45			6,915	1.47		
Deviation	2,429	0.12			2,707	0.11		
Production Rates With New Regulatory Controls and with Water and PRC Programs								
Total Tonnes for Period	5,245,372		30%	32%	5,231,567		56%	59%
Average	7,680	1.51			7,682	1.51		
Deviation	1,901	0.11			1,903	0.11		
Composite Production Rates	16,281,245	Production Increase 9.5%			8,924,291	Production Increase 12.2%		

Table 1 – Early Project Performance Results

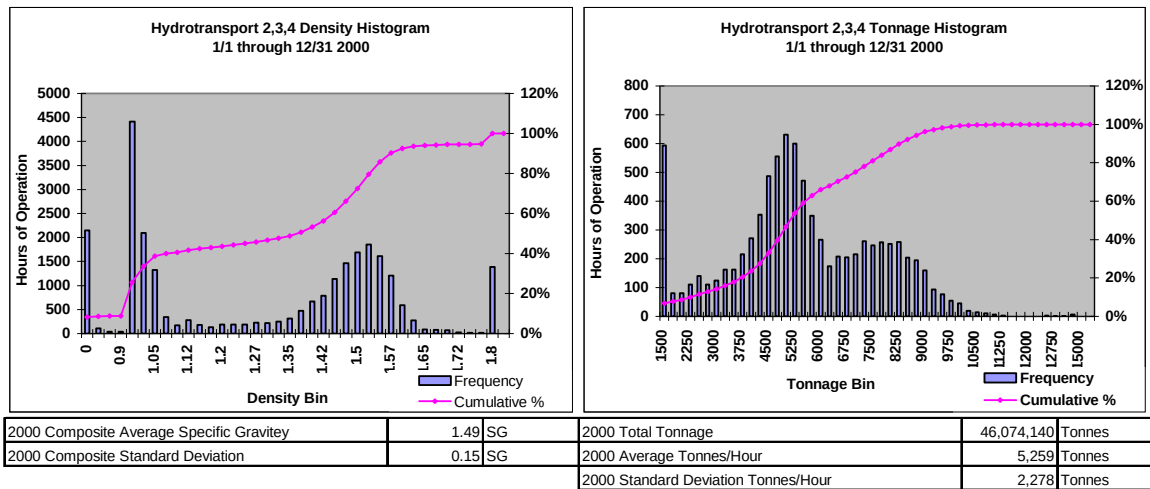


Figure 5a – 2000 Density and Production

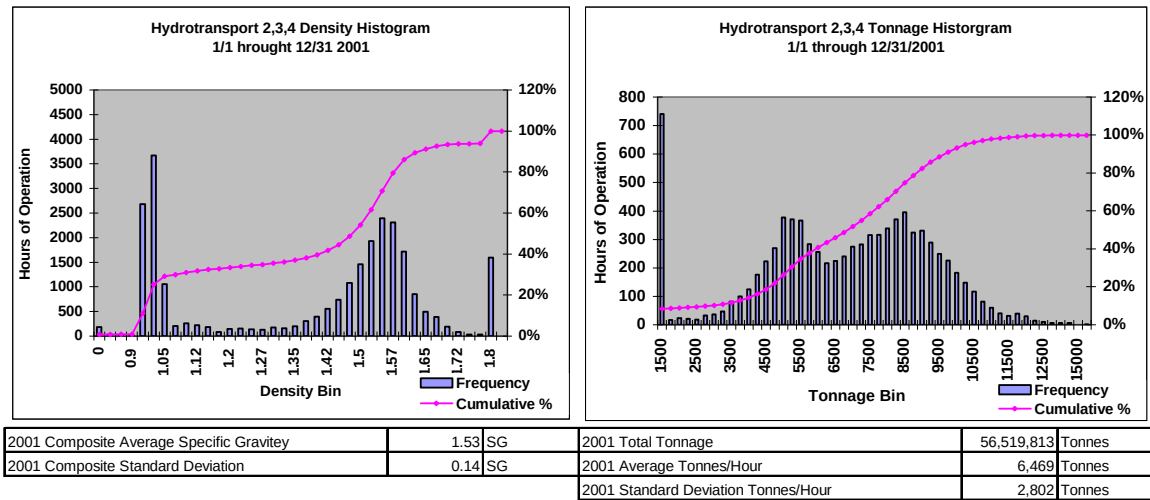


Figure 5b – 2001 Density and Production

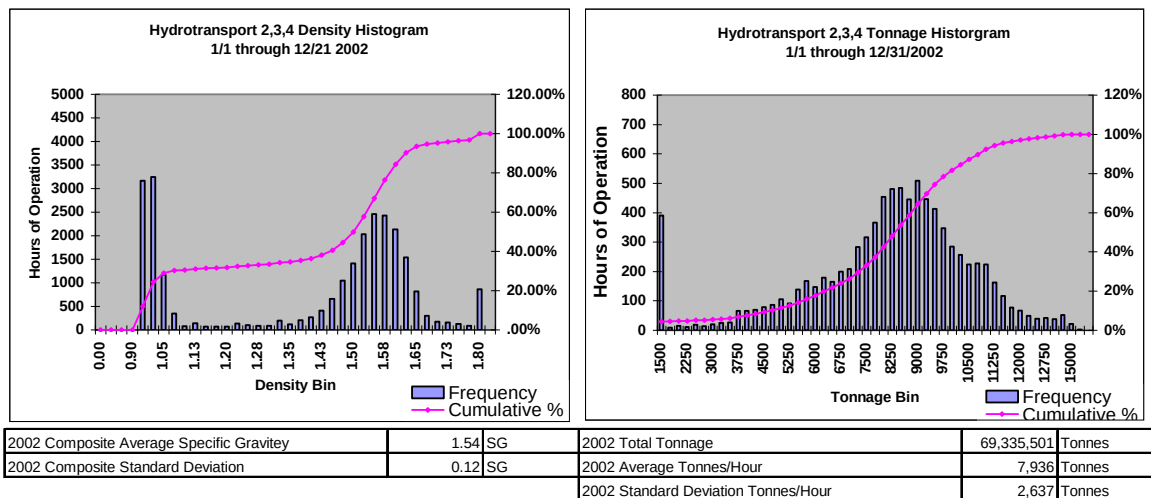


Figure 5c – 2002 Density and Production

CONCLUSION:

This advanced control project is halfway complete and exceeded the original financial objectives by a factor of two. The following are project implementation observations:

- Critical systems were operating on the performance edge where control errors would result in instability and lost production. Normal engineering and commissioning estimates cannot be used in that type of environment. The engineering and programming effort increased by a factor of three to apply methods to safely integrate into the existing systems, assure a backup mechanism through the existing regulatory controls, test control strategies, and finally integrate into the existing systems without causing production outages.
- From the business perspective, Suncor Energy is a unique company. The organization is empowered to develop incentives to drive towards a successful project. In this case, performance guarantees were fairly administered to provide the necessary financial incentives to assure a focused project team.
- Integration of the Advanced Control Project with ongoing capital projects forms an interdependence that makes individual performance benefits difficult to assign. The simplest method was to turn the

controls off during the commissioning stage and measure the results as given in Table 1- *Early Project Performance Results*. This returned a 9% to 12% production increase based on the measurement method. But, Table 1 – *Overall Project Summary* shows the two-year overall project production increase of 22.7%/year. Those rate improvements could not be reached without the enabling control technology but the process modifications provided the opportunity. Eventually, the measurement of overall gain remains beneficial but the effort of assigning individual gains reaches a point of diminishing returns.

- The project team continues to seek opportunities of “Infinite Return”. This is the application of low cost methods and technologies such as operating procedure changes, software applications or improved sensors that result in large benefits at minimal cost. Unfortunately, sometimes implementation of “low cost solutions” can become surprisingly expensive. A specific example is using non-secure communication systems for control data resulting in project delays and production losses.
- Leveraging team efforts benefited multiple organizations. An example is the documentation effort that has reduced the time for finding and communicating control information time by 25%. This information was formatted to the engineering, operating and training packages.

Production Run	2000	2001	2000/2001	2002	2001/2002	2000	Engineering Units
Process Parameter			Change		Change	Baseline Gain	
Production							
Annual Ore Processing	46.1	56.5	22.7%	69.3	22.7%	50%	Million Metric Tonnes/Year
Average Tonnage	5259	6469	23.0%	7936	22.7%	51%	Tonnes/Hour
Tonnage Deviation	2278	2802	23.0%	2637	-5.9%	16%	Tonnes/Hour
Production Increase			\$104.00		\$140.00	\$244.00	C\$MM/year
Energy Utilization							
Hot Process Water 200F	141.5	122.3	-13.6%	117.8	-3.7%	-17%	Gallons/Tonne
Process Water Deviation	0.15	0.14	-6.7%	0.12	-14.3%	-20%	Gallons/Tonne
Energy Savings			\$8.70		\$2.50	\$11.20	C\$MM/year

Table 2 - Overall Project Performance Results

REFERENCES

¹ Gray, Murray, Masliyah, Jacob: University of Alberta – *Extraction and Upgrading of Oilsands Bitumen* – Short Course May 2000, Page 1-1

² Gray, Murray, Masliyah, Jacob: University of Alberta – *Extraction and Upgrading of Oilsands Bitumen* – Short Course May 2000, Page 4-1

³ Wood, Robinson – *The Powerhouse Control System Audit as a Means of Insuring a Successful System Design and Implementation*, IETC 1998.

⁴ Robinson - *Advanced Process Management and Implementation*, IETC 1999.